MATHEMATICS

Applications and interpretation

Higher Level | Set B | Paper 2

Challenge mock examination

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A graphic display calculator and a clean Mathematics: applications and interpretation formula booklet are required.

• Answer all questions. Show working and explain mathematical reasoning in context.

• All questions are compulsory. There is no optional section.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures, unless a question specifies a different accuracy.

• Keep unrounded values in subsequent calculations. State units where relevant.

• Diagrams are not necessarily drawn to scale. Use the stated data and conditions.

• The number in square brackets gives the maximum mark for a part.

Thriving Scholars original practice | 2026 challenge series

Question 1 | Question paper

Question 1 [Maximum mark: 18]

A community centre can buy an energy system for \$18000 now. It produces savings of \$3200 at the end of year 1; these savings increase by 3% each year for eight years. Maintenance of \$900 is paid at the end of each year. A resale receipt of \$2000 is received at the end of year 8 in addition to that year's savings. All amounts quoted are nominal cash flows. Define net present value as discounted receipts minus discounted payments.

(a) Write an expression for the net present value $V(r)$ when the effective annual discount rate is r. [4]

(b) Find the net present value at $r = 0.06$, to the nearest dollar, and interpret its sign. [2]

(c) The internal rate of return is the positive value of r for which $V(r) = 0$. Find it as a percentage. Explain why there is at most one positive solution for these cash flows. [4]

(d) Annual inflation is 2%. Find the annual growth rate in the purchasing power of the gross savings. [3]

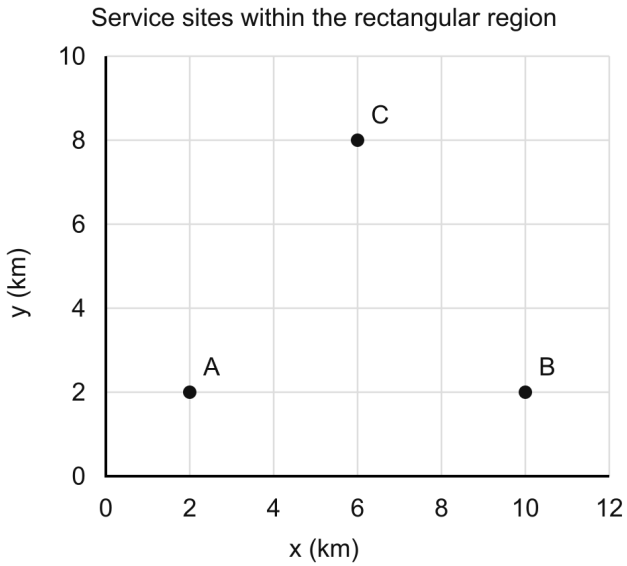
(e) An alternative system costs the same initially and gives a guaranteed net receipt of \$2600 at the end of each of nine years, with no resale value. Compare the two systems using a 6% discount rate. [3]

(f) State two distinct forecasting risks in the first system which the NPV calculation does not remove. [2]

Question 2 | Question paper

Question 2 [Maximum mark: 18]

Three service centres have coordinates $A(2, 2)$, $B(10, 2)$ and $C(6, 8)$, in kilometres. Their service region is the rectangle $0 \leq x \leq 12$, $0 \leq y \leq 10$. Initially each location is allocated to its nearest centre by straight-line distance. The sites are shown on the diagram.



(a) Find the perpendicular bisector of AB. [2]

(b) Find the perpendicular bisector of AC and the point where the three Voronoi boundaries meet. Verify that this point is also equidistant from B and C. [5]

(c) Find the area of the Voronoi cell allocated to A within the rectangular region. [4]

(d) Within the cell for A, locate the point farthest from A and find its distance. You may use the fact that the maximum distance from a fixed point over a convex polygon is attained at a vertex. [4]

Question 2 | Question paper

(e) At $D(5, 4)$, straight-line response speeds from A, B, C are respectively $40, 80, 30$ km per hour. Determine the fastest centre and explain why this may differ from the Voronoi allocation. [3]

Question 3 [Maximum mark: 19]

In a clinic, hourly counts of appointments and walk-ins are independent Poisson variables with means 2.4 and 3.6. Thus the total hourly count N is Poisson with mean 6. Counts in disjoint time intervals are independent, with constant rates. In parts (d) and (e), use one-tailed tests for an increase in the total arrival rate.

(a) Find the probability of at least ten arrivals in one hour. [3]

(b) Given that there are five arrivals in an hour, the number of appointments has a binomial distribution. Find its success probability and calculate the probability that at least three of the five are appointments. [4]

(c) A two-hour session has capacity for m arrivals. Find the smallest integer m for which the probability of arrivals exceeding capacity is less than 0.01. Show the adjacent probabilities used. [3]

(d) In eight observed hours there are 65 arrivals. Test at the 5% level whether the hourly total rate has increased from 6. State hypotheses, distribution and conclusion. [4]

(e) For the same eight-hour test, find the rejection region with the largest Type I error probability strictly below 0.05. Then calculate and interpret the probability of a Type II error if the true hourly rate is 7. [5]

Question 4 [Maximum mark: 19]

A learning website has pages A, B, C. Each minute a visitor either stays or follows one link. Using column state vectors, the transition matrix is

$$T = \begin{pmatrix} 0.6 & 0.2 & 0.1 \\ 0.3 & 0.5 & 0.3 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}$$

, where T_{ij} is the probability of moving from page j to page i. Initially 1000 visitors are all on A. Each visitor remains in the model and evolves independently. Repeated visits to C in different minutes are counted separately.

(a) State the probability of moving from B to C in one minute and find the expected state vector after one minute. [2]

(b) Find the expected numbers on each page after two minutes. [3]

(c) Find the steady-state proportions on the three pages. Show the equations and normalization used. [5]

(d) Explain why the steady state is approached from any initial probability distribution. [2]

(e) Find the expected total number of visits to C during minutes $1, 2, 3, 4, 5$ by the 1000 visitors. [4]

(f) An additional link changes the probabilities from page C to $(0.25, 0.30, 0.45)^T$, leaving the other columns unchanged. Find the new steady state and assess whether this increases the long-run fraction on C. [3]

Question 5 [Maximum mark: 18]

A vehicle is tested on one dry road. Speed v is in km per hour and braking distance d is in metres. Use a power model $d = kv^b$, fitted by linear regression of $\ln d$ on $\ln v$.

v	20	30	40	50	60	70	80
d	3.2	6.9	12.6	20.1	29.5	39.8	53.0

(a) Find k and b, and the correlation coefficient for the linearized data. [4]

(b) Use the model to predict the braking distance at 65 km per hour and find the residual at 80 km per hour, taking residual as observed minus predicted distance. [3]

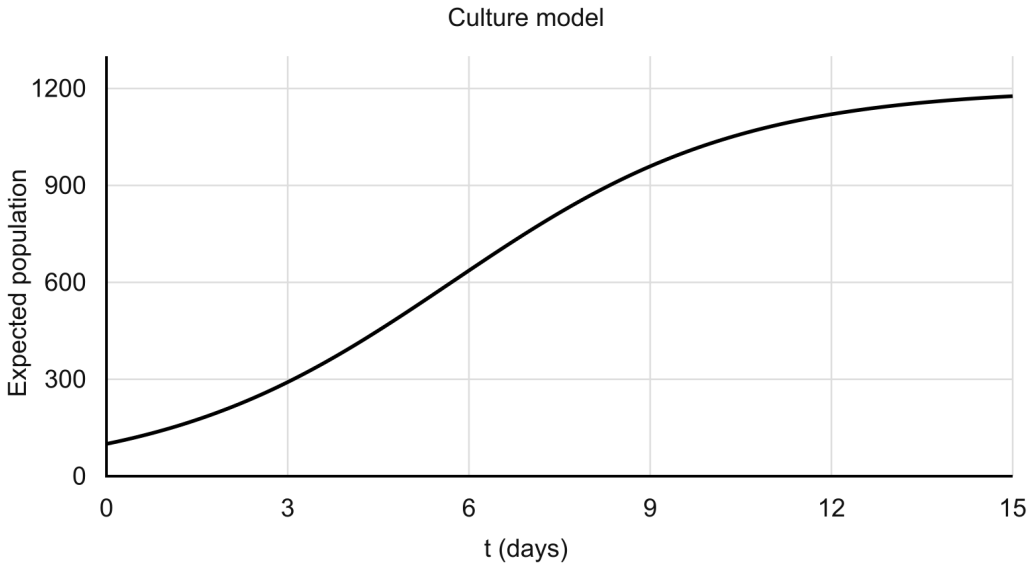
(c) The driver's reaction time is 1.1 seconds. Find the largest speed for which the modelled total stopping distance is at most 50 metres. Show the equation solved and justify choosing the boundary solution. [4]

(d) An alert system reduces reaction time to 0.85 seconds without changing braking performance. Find the new largest speed for the same 50-metre stopping distance and explain why the braking coefficient k is unchanged. [4]

(e) Explain why the high linearized correlation does not justify using this model on an icy road at 120 km per hour. State what quantity the log-linear regression minimizes. [3]

Question 6 [Maximum mark: 18]

A managed culture is modelled by $N(t) = \dfrac{1200}{1 + 11e^{-0.42t}}$, where N is the expected number of organisms and $t \geq 0$ is time in days. A proposed differential model is $N\prime = rN(1 - N/1200)$. Resource conditions are held fixed during the observation period.



(a) Find the initial population and limiting population, and interpret the constant 1200. [3]

(b) Differentiate $N(t)$ and show that the proposed differential model holds for a particular value of r. [4]

(c) Find the largest instantaneous growth rate and the time at which it occurs. Justify that it is a maximum. [4]

(d) Find the mean population over days 5 to 10. Explain why this need not equal the population on day 7.5. [4]

Question 6 | Question paper

(e) Explain why this model predicts no overshoot above 1200 from the stated initial condition. Give one plausible change to the real culture which would invalidate the fixed carrying-capacity assumption. [3]

End of question paper

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Worked solution book

Complete this paper before consulting the worked solutions. Each part has method (M), accuracy (A) and reasoning (R) points. The number following a code is its mark value. Award only the points supported by the candidate's retained working.

Accept mathematically equivalent valid methods. Numerical answers may use technology, but should be supported by equations, test definitions, bounds, labelled graphs or other appropriate mathematical working. Retain unrounded intermediate values.

Use the portal to enter a whole-number self-assessed mark for each part. These are independent practice papers; the self-assessed report does not calculate an official IB grade.

Question 1 [Maximum mark: 18]

(a) [4 marks]

M1 The net annual saving in year j is $3200(1.03)^{j-1} - 900$.

M1 Discount each year-end flow by $(1+r)^j$.

A1 Include $2000/(1+r)^8$ for the additional resale receipt.

A1 $V(r) = -18000 + \sum_{j=1}^{8} \{3200(1.03)^{j-1} - 900\}/(1+r)^j + 2000/(1+r)^8$.

(b) [2 marks]

A1 $V(0.06) = -444.6266$ dollars, giving -445 dollars to the nearest dollar.

R1 The negative value means the investment does not meet a 6% effective annual required return under these forecasts.

(c) [4 marks]

M1 Solve the stated NPV equation numerically for $r > 0$.

A1 $r = 0.05434374$, or 5.434374% per year.

R1 Every future net receipt is positive.

R1 Each discounted positive receipt decreases as r increases, so V is strictly decreasing for $r > 0$.

(d) [3 marks]

M1 Divide the nominal savings growth factor by the inflation growth factor.

A1 The real growth factor is $1.03/1.02$.

A1 The real growth rate is 0.9803922% per year.

(e) [3 marks]

M1 For the alternative, calculate $-18000 + \sum_{j=1}^{9} 2600/(1.06)^j$.

A1 The alternative NPV is -315.6001 dollars.

R1 Compare this value with -444.6266 for the first system; choose the higher NPV if one must be selected. A negative NPV still fails the required return.

(f) [2 marks]

R1 The actual future savings or their growth may differ from the forecast, for example because energy prices change.

R1 Maintenance, service life or resale value may differ; discounting does not guarantee these uncertain cash flows.

Question 2 [Maximum mark: 18]

(a) [2 marks]

M1 AB is horizontal with midpoint $(6, 2)$.

A1 The bisector is $x = 6$.

(b) [5 marks]

M1 Equate $(x-2)^2 + (y-2)^2$ and $(x-6)^2 + (y-8)^2$.

A1 The bisector is $2x + 3y = 23$.

M1 Intersect this with $x = 6$.

A1 The common vertex is $(6, 11/3)$.

R1 The BC bisector is $-2x + 3y = -1$, satisfied by the same point.

(c) [4 marks]

M1 The cell satisfies $x \leq 6$ and $2x + 3y \leq 23$, with the rectangle bounds.

A1 Its vertices are $(0, 0), (6, 0), (6, 11/3), (0, 23/3)$.

M1 Use a trapezium area or integrate the upper edge from $x = 0$ to 6.

A1 The area is $34\ \text{km}^2$.

(d) [4 marks]

M1 Compare distances from A to the four cell vertices.

A1 The squared distances are $8, 20, 169/9, 325/9$, respectively.

A1 The farthest vertex is $(0, 23/3)$.

A1 The maximum distance is $\sqrt{325}/3 = 6.009252\ \text{km}$.

(e) [3 marks]

M1 Distances from D to A, B, C are $\sqrt{13}, \sqrt{29}, \sqrt{17}$ km; divide by the respective speeds.

A1 Times are 5.408327, 4.038874 and 8.246211 minutes, so B is fastest.

R1 Distance-only allocation gives A, but unequal speeds mean nearest distance need not minimize response time.

Question 3 | Worked solutions

Question 3 [Maximum mark: 19]

(a) [3 marks]

M1 $P(N \geq 10) = 1 - P(N \leq 9)$, with $N \sim \text{Po}(6)$.

A1 Use the Poisson cumulative probability with upper included count 9.

A1 The probability is 0.08392402.

(b) [4 marks]

R1 The appointment fraction is $2.4/(2.4 + 3.6) = 0.4$.

A1 The conditional count is $\text{Bin}(5, 0.4)$.

M1 Use $1 - P(X \leq 2)$.

A1 The conditional probability is 0.31744.

(c) [3 marks]

M1 The two-hour count is $\text{Po}(12)$; require $P(N > m) < 0.01$.

A1 At $m = 21$ the tail is 0.006065147; at one less it is 0.01159774.

A1 The smallest capacity is 21.

(d) [4 marks]

A1 $H_0 : \lambda = 6$ per hour, $H_1 : \lambda > 6$ per hour.

M1 Under H_0, the eight-hour count is $X \sim \text{Po}(48)$.

A1 $p = P(X \geq 65) = 0.01117098$.

R1 Reject H_0 since $p < 0.05$; the observed count supports an increased rate.

(e) [5 marks]

M1 Find the smallest integer k with $P(\text{Po}(48) \geq k) < 0.05$.

A1 Reject for $X \geq 61$; the actual Type I error probability is 0.03950011.

M1 At rate 7, the eight-hour count is $\text{Po}(56)$; a Type II error is $X < 61$.

A1 The Type II probability is 0.7309104.

R1 This is the probability that the test fails to detect an increase when the true hourly rate is 7, not the probability that the null hypothesis is true.

Question 4 | Worked solutions

Question 4 [Maximum mark: 19]

(a) [2 marks]

A1 The probability from B to C is the row C, column B entry, 0.3.

A1 The expected counts are $(600, 300, 100)^T$.

(b) [3 marks]

M1 Use $T^2(1000, 0, 0)^T$.

A1 The probability vector is $(0.43, 0.36, 0.21)^T$.

A1 The expected counts are $(430, 360, 210)^T$.

(c) [5 marks]

M1 Solve $T\mathbf{s} = \mathbf{s}$.

A1 For example, $0.4a = 0.2b + 0.1c$ and $0.5b = 0.3a + 0.3c$.

M1 Add $a + b + c = 1$ to give a unique normalized state.

A1 The steady state is $(0.275, 0.375, 0.35)^T$.

R1 These are long-run fractions, and multiplying by 1000 gives the corresponding limiting expected counts.

(d) [2 marks]

R1 Every entry of T is strictly positive, so the transition matrix is regular.

R1 A regular Markov chain approaches its unique steady-state distribution independently of its initial distribution.

(e) [4 marks]

M1 For one visitor initially on A, use the C component of $T^n(1, 0, 0)^T$ in minute n.

M1 Add those five probabilities, rather than using only the minute-5 occupancy.

A1 The expected visits per visitor total 1.23123.

A1 The expected aggregate is 1231.23 visits; repeated visits are included.

Question 4 | Worked solutions

(f) [3 marks]

M1 Replace the third column and solve $T_{\text{new}}\mathbf{s} = \mathbf{s}$, $a + b + c = 1$.

A1 The new proportions are $(0.3557692, 0.375, 0.2692308)^T$.

R1 The fraction on C decreases from 0.35 to 0.2692308, so the change does not meet that objective.

Question 5 [Maximum mark: 18]

(a) [4 marks]

M1 Fit $\ln d = a + b \ln v$.

A1 $a = -4.964212$ and $b = 2.036363$.

A1 $k = e^a = 0.00698345$.

A1 The correlation coefficient of the log data is $r = 0.9998259$.

(b) [3 marks]

M1 Substitute the speeds into kv^b, using unrounded regression coefficients.

A1 The prediction at 65 is 34.34158 m.

A1 At 80, the residual is $53 - 52.41465 = 0.5853497$ m.

(c) [4 marks]

M1 The speed in metres per second is $v/3.6$, so total stopping distance is $1.1v/3.6 + kv^b$.

M1 Solve $1.1v/3.6 + kv^b = 50$ on the tested speed interval.

A1 The boundary speed is 61.89802 km per hour.

R1 Both terms increase for $v > 0$ since $k, b > 0$, so this is the largest admissible speed.

(d) [4 marks]

M1 Replace only the reaction term: $0.85v/3.6 + kv^b = 50$.

A1 The new boundary speed is 65.23683 km per hour.

R1 The system changes reaction time, not the measured relationship between speed and braking distance.

R1 Therefore the original braking regression and its coefficient k remain in the model.

(e) [3 marks]

R1 The road condition differs from that of the measurements, so the same braking parameters need not apply.

R1 120 lies outside the observed speed range, making this an extrapolation.

R1 The fitted line minimizes the sum of squared residuals in $\ln d$, not in the original distances d.

Question 6 [Maximum mark: 18]

(a) [3 marks]

A1 $N(0) = 1200/12 = 100$.

A1 $N(t)$ approaches 1200 as t increases.

R1 1200 is the modelled carrying capacity under the fixed resource conditions.

(b) [4 marks]

M1 Apply the chain rule to $1200(1 + 11e^{-0.42t})^{-1}$.

A1 $N\prime = 1200(11)(0.42)e^{-0.42t}/(1 + 11e^{-0.42t})^2$.

M1 $1 - N/1200 = 11e^{-0.42t}/(1 + 11e^{-0.42t})$.

A1 The expressions agree for $r = 0.42$ per day.

(c) [4 marks]

M1 Treat growth as $G(N) = 0.42N(1 - N/1200)$.

R1 This downward-opening quadratic is maximized at $N = 600$.

A1 The maximum rate is 126 organisms per day.

A1 $11e^{-0.42t} = 1$, giving $t = 5.709274$ days.

(d) [4 marks]

M1 The time mean is $\frac{1}{5}\int_5^{10} N(t)\,dt$.

A1 The mean is 799.7329 organisms.

A1 The midpoint population is $N(7.5) = 815.5646$.

R1 For a nonlinear function, an interval average generally differs from the value at the midpoint; they answer different questions.

(e) [3 marks]

R1 The denominator is $1 + 11e^{-0.42t} > 1$ for every finite $t \geq 0$, so $N(t) < 1200$.

R1 Also $N\prime > 0$ while $0 < N < 1200$, with growth tending to zero near capacity.

R1 A change in nutrient supply, available space, temperature or waste accumulation could change the effective carrying capacity.